BME 201 Fall 2026
PA6: Stress-Strain Analysis (AI is permitted)
Due: Friday, 10/23, 5 pm

Background

Stress-strain curves give information about the mechanical properties of a metal. Knowing the stress a material can tolerate before deforming or failing is a critical factor when designing a component. Akin to the critical stresses are the resilience and toughness, which respectively define how much energy the material can absorb before deforming permanently or failing entirely.

Stress-strain curves can be measured with a tension test in a load frame, where a dogbone-shaped specimen is stretched uniaxially until it fractures. The load frame has a displacement sensor to measure how far the sample has stretched from its initial position and a load cell measures the force required to maintain that displacement. “Force” and “displacement” depend on the dimensions of the sample and are thus very specific to a sample. To generalize to all geometries of the same material, force and displacement must be converted to the equivalent material properties: stress and strain.

Problem

You have run a tension test on a sample of aluminum and saved your force-displacement data in *aluminum.xlsx*. The first column is the displacement (m), which is zeroed before the test. The second column is the tensile force (N) applied by the load cell to maintain the corresponding displacement. Your sample has a gage length of 38 mm and a circular cross-section with radius of 1.5 mm.

Write a program that imports the data and extracts the following properties:

- Proportional stress and corresponding strain
- Yield stress and corresponding strain using the 0.2% offset rule
- Ultimate stress and corresponding strain
- Fracture stress and corresponding strain
- Elastic modulus
- Stress to hundredths place at a strain of EXACTLY 0.1000. Make sure that this strain is declared as a variable so that we can adjust to find any strain, and make sure that your `fprintf` displays both the value of strain and that of the corresponding stress.

Print the values to the command window with `fprintf` with appropriate precision. Plot stress vs. strain and show some type of mark at the proportional, yield, ultimate, and fracture points.

Stress and elastic modulus are in Pa, but make sure to use an appropriate metric prefix (e.g.,μPa, TPa) to keep the notation clean. For example, instead of 0.00005421 Pa, you should report 54.21 μPa. For strain, report your values to the nearest ten thousandths place. No metric prefixes are required for strain.

Your code should be able to handle another similar data set (i.e., ductile metal, not an elastomer or very brittle material) but with a different force-displacement data set.

Scoring

Doing everything above will earn an 80/100. For a higher grade, your code should also do the following:

- (5 pts) Label the critical points on your graph as *Proportional*, *Yield*, *Ultimate*, and *Fracture*. Make sure it works with different data sets and that the writing doesn't cover the data points.
- (5 pts) Calculate the resilience and toughness in MJ/m^3 and print them to the screen. Look at the numerical integration functions or devise your own method. We will cover this later.
- (5 pts) Shade the area under the curve in red corresponding to the elastic region. Shade the area under the curve in cyan corresponding to plastic region.
- (5 pts) Smooth your force data before converting it to stress with a 25-point averaging filter.

Theory

Stress (σ) and strain (ε) are analogous to force (F) and displacement (Δl), respectively:

$$\sigma = \frac{F}{A} \quad \text{and} \quad \varepsilon = \frac{\Delta l}{l}$$

where A is the cross-sectional area and l is the specimen gage length.

A typical stress-strain curve has a linear region at the beginning followed by a non-linear region that increases slowly to a maximum, then decreases until fracture. The slope of the linear region is the **elastic modulus, *E***. The end of the linear regime is called the **proportional limit**, and it has (x,y) coordinates of (ε_p, σ_p).

Unfortunately, the proportional limit is difficult to detect exactly; instead, the **yield point** is often used as the end of the linear region. Its coordinates are (ε_y, σ_y). Starting along the x axis at a strain value of $\varepsilon = 0.002 = 0.2\%$ and following a line with a slope of E, the intersection of this line with the stress-strain curve is the yield point. Any stress exceeding the yield stress will cause permanent, or plastic, deformation; stresses lower than the yield stress cause non-permanent, or elastic, deformation that is recovered upon unloading.

Continuing past the yield point, the stress reaches a maximum value at its **ultimate point**. This point has coordinates of (ε_u, σ_u). Beyond the ultimate point, the stress decreases until the material fails at its **fracture point**, which has coordinates of (ε_f, σ_f).

The SI units of stress are pascals, Pa, which are N/m^2. This is equivalent to one unit of energy per volume, or J/m^3. Thus, the area under the stress-strain curve has units of J/m^3. The **resilience** is the amount of energy a material can tolerate before it yields, while the **toughness** is the total energy absorbed before fracture.



Submission

Submit your main program.

Submit a summary as a PDF of how you used AI. Use bullets and specific line numbers for clarity. Any functions not discussed in class or zyBooks MUST be explained here. There is **no** penalty for documented AI use, but undocumented AI use is not allowed. If you did **not** use AI, you must still submit a PDF saying as much.